Type MS · Connecting to the internet: switch, WAP, router and modem

Answer 1 of 3 · Question 1 part d · 3 marks

2025 JUN · Paper 01 · extract 1 of 4 · find part d in this mark scheme extract

<table>
<tr><td>1(a)</td><td>Award one mark for each correct row:

<table><tr><th>Statement</th><th>DNS</th><th>PAN</th><th>WAN</th><th>WWW</th></tr><tr><td>A network of networks not limited to a single location</td><td></td><td></td><td>✓</td><td></td></tr><tr><td>A network that connects devices within the range of a single user</td><td></td><td>✓</td><td></td><td></td></tr><tr><td>A set of hypertext documents accessed through the HTTP protocol</td><td></td><td></td><td></td><td>✓</td></tr><tr><td>A system that maps domain names to IP addresses</td><td>✓</td><td></td><td></td><td></td></tr></table></td><td>Do not award a mark for a row containing more than one tick.</td><td>4</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(b)</td><td>Award one mark for each of:

• Transport (layer) (1)
• (Data) link / network link / network interface (layer) (1)</td><td>Accept
• transfer or transmission for Transport

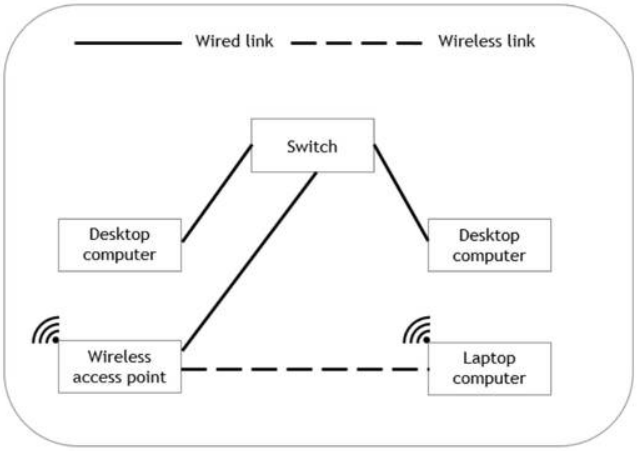
Do not accept:
• Network layer and Application layer (in the question)
• Data access layer, Internet layer (alternative names for network layer)
• Data’ or ‘network’ without ‘link’ or ‘interface’</td><td>2</td></tr>
<tr><th>Question</th><th>Answer</th><th>Additional Guidance</th><th>Mark</th></tr>
<tr><td>1(c)</td><td>The only correct answer is D

A is not correct because HTTP is used to transfer hypertext
B is not correct because it is used to view email held on a server
C is not correct because it is used to download email from a server</td><td></td><td>1</td></tr>
</table>

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Question	Answer	Additional Guidance	Mark
1(d)	Award **one** mark for each correct connection: • One solid line from wireless access point to switch (1) • One solid line from each desktop to switch (1) • One dashed line from laptop computer to wireless access point (1) 	No marks should be awarded if any additional lines have been drawn for each mark point. Do not double penalise (e.g. if you have penalised a second line from desktop to WAP then don't penalise extra line from WAP to desktop)	3

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Question	Answer	Additional Guidance	Mark
1(e)	Award up to **two** marks for a linked description such as: • To find weaknesses in a system (1) fix/repair weakness (1) • To find vulnerabilities in a system (1) by testing the effectiveness of the security measures in place (1) • An ethical hacking attack on a network (1) to identify vulnerabilities / recommend measures/take action to improve security (1) • Tests whether users of the network (1) can be tricked into giving away information / giving criminals access to the network / downloading malware (1) • To test the system security (1) / sanctioned by the owner (1)	Explanation can be a combination of different marking points. E.g. MP1a and MP2b	2
		Total for question 1	**12**

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Answer 1 of 3 · Question 1 part d · 3 marks

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Question	Answer	Additional Guidance	Mark

Type MS · Connecting to the internet: switch, WAP, router and modem

Answer 2 of 3 · Question 2 part d · 6 marks

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2(a)	Award **one** from each: Zafer: 1. Choose strong (1) authentication credentials (username/password) 2. Not sharing his credentials (1) 3. Not having automatic login to cloud / leaving his machine unattended (1) 4. Changing password regularly (1) 5. Password protect documents (1) Provider: 1. Infrastructure (e.g. firewall/servers) must be secure from unauthorised access (1) 2. Keeping their security software up to date (1) 3. Policy and procedures effective in preventing an insider attack / a data breach (1) 4. Data protection laws in the resident country must be obeyed (1) 5. Backup and restore procedures (1) 6. Encryption (1)		2

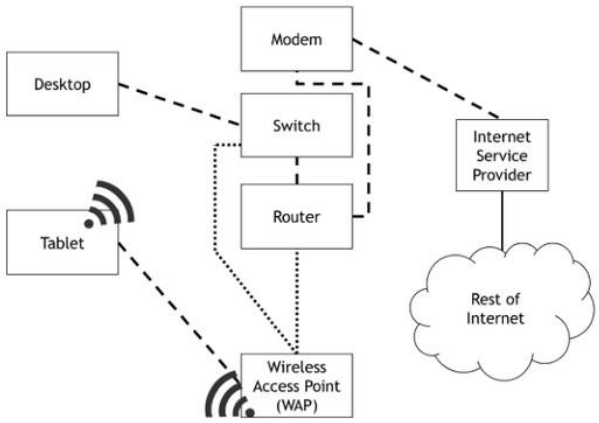
Question Number	Answer	Additional Guidance	Mark
2(b)	**Two** marks if completely correct. **One** mark for any two correct cells. **Robert** / **Zafer** Washing machine design drawings: Read / Read A folder of manuals for new machines: None / Read, Write Or Write, Read	Ignore spelling mistakes	2

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Answer 2 of 3 · Question 2 part d · 6 marks

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Question Number	Answer	Additional Guidance	Mark
2(c)	Award **one** mark from each category (maximum of 2) Input: • www.pearson.com / website domain (1) • Uniform Resource Locator / URL (of Pearson website) (1) Output: • 2.20.38.113 (1) • Pearson's machine IP(v4) address (1)	Don't penalise spelling or typos, especially in addresses	2

Question Number	Answer	Additional Guidance	Mark
2(d)	One mark for each dotted connection, correctly located. Maximum of 6 marks. Note: WAP can be connected to either the Switch or the Router, but not both  		6

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Answer 2 of 3 · Question 2 part d · 6 marks

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Question Number	Answer	Additional Guidance	Mark

Type MS · Connecting to the internet: switch, WAP, router and modem

Answer 3 of 3 · Question 4 part d · 3 marks

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4(a)	**The only correct answer is C** ***A*** *is not correct because* ***mebibits*** *is a measure of storage capacity* ***B*** *is not correct because* ***mebibytes*** *is a measure of storage capacity* ***D*** *is not correct because transmission speeds are measured only in bits per second, not* ***bytes per second***		1

Question Number	Answer	Additional Guidance	Mark
4(b)	Award **one** mark for each correct cell: **Type** / **Example** / **Number of bits** IPv4 / 192.169.0.3 / 32 (1) IPv6 / 1050:a500:00c0:0440:0006:0300:700d:436f / 128 (1)		2

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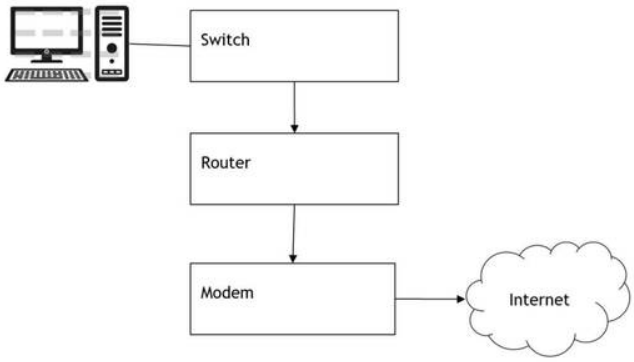
Question Number	Answer	Additional Guidance	Mark
4(c)(i)	Award up to **two** marks for a linked description, such as: • In a wired network packets travel over (copper/fibre-optic) cables (1), whereas in a wireless network packets travel over the air (1)	Allow radio signals, infra-red, microwave for air (must be more than just waves) Allow twisted pair, CAT for copper. Allow electrical/light signals (not just electricity)	2

Question Number	Answer	Additional Guidance	Mark
4(c)(ii)	Award **one** mark for any of the following up to a maximum of **two** marks: 1. A protocol (suite/group/family) (1) 2. Used on a wired network / wired connection (1) 3. Defines physical parts/type of cable(twisted pair, CAT6)/type of connector (1) 4. Defines how packets are checked for errors (1) 5. Defines the speed of transmission (1) 6. Operates at the link layer of the TCP/IP stack (1)		2

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Answer 3 of 3 · Question 4 part d · 3 marks

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Question Number	Answer	Additional Guidance	Mark
4(d)	Award **three** marks for all of switch, router and modem in the correct order as in diagram Award **two** marks for all of switch, router and modem in any order Award **one** mark for any of switch or router or modem in any box 		3

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Answer 3 of 3 · Question 4 part d · 3 marks

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Question Number	Answer	Additional Guidance	Mark
4(e)	Award up to **two** marks for a linked description, such as: Validation is used to check that the data/password entered by a user is fit for processing/meets specified requirements (1), whereas authentication is used to verify the identity of the user/prove that users are who they say they are (1)	For validation 'specified requirements' a good description of the check can be given e.g. presence check, length check, format check	2

Question Number	Answer	Additional Guidance	Mark